

# Generative Reverse CTMC for the Moran Model: The Independent Pseudo-Marginal Approximation

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## 1 Forward Continuous-Time Markov Chain (CTMC)

The forward particle system operates on a state space defined by  $N$  particles moving on a discrete spatial grid. The joint state is  $X = (x^{(1)}, \dots, x^{(N)})$ . The forward process incorporates two competing dynamics: spatial diffusion (mutation) and localized copying (resampling).

The forward transition rate matrix  $Q(t)$  drives the system. For a state  $X$ , the rate to transition to an adjacent state  $Y$  is:

1. **Mutation (Independent Random Walk):** A single particle  $i$  takes a spatial step  $dx \in \{(0, 1), (0, -1), (1, 0), (-1, 0)\}$ . The resulting state  $Y$  has  $y^{(i)} = x^{(i)} + dx$ . The forward rate is:

$$Q_{X \rightarrow Y}(t) = \gamma_{\text{mut}} \lambda(t)$$

where  $\lambda(t)$  is the time-scaling factor (e.g.,  $1/(1-t)$ ) ensuring convergence to the uniform stationary distribution as  $t \rightarrow 1$ .

2. **Resampling (Interacting Coping):** A particle  $i$  spontaneously copies the spatial coordinate of another particle  $j$ . The resulting state  $Y$  has  $y^{(i)} = x^{(j)}$ . The forward rate is driven by an RBF spatial kernel  $K$ :

$$Q_{X \rightarrow Y}(t) = \frac{\kappa}{N} K(x^{(i)}, x^{(j)}) \lambda(t)$$

## 2 The Reverse Generative CTMC

To use this process generatively, we must simulate the exact reverse-time CTMC operating from  $t = 1$  backward to  $t = 0$ . According to Anderson's theorem (1982), the reverse-time transition rates from state  $X$  to general state  $Y$  are analytically given by:

$$R_{X \rightarrow Y}(t) = Q_{Y \rightarrow X}(t) \frac{p_t(Y)}{p_t(X)}$$

where  $p_t$  is the unconditional marginal probability of the joint distribution at time  $t$ .

In generative modeling, we condition on the ground truth target  $X_0$ . Given a neural network prediction  $\hat{X}_0$ , we compute the ratio using the conditional transition probabilities:

$$\frac{p_t(Y)}{p_t(X)} \approx \frac{P(X_t = Y \mid X_0 = \hat{X}_0)}{P(X_t = X \mid X_0 = \hat{X}_0)}$$

## 2.1 The Intractability Bottleneck

Because the Moran model explicitly contains resampling ( $\kappa > 0$ ), the particles interact. For an interacting many-body system, the state transitions do not factorize, meaning  $P(X_t | X_0)$  involves solving the Master Equation over  $N$  strongly coupled particles. This is mathematically intractable. Consequently, the true generative reverse rates  $R_{X \rightarrow Y}(t)$  cannot be exactly calculated.

## 3 Option A: The Independent Pseudo-Marginal Approximation

To bypass this intractability without destroying the interacting generative properties of the model, we employ an analytical approximation strictly during the evaluation of the conditional ratio.

We assume  $\kappa = 0$  *only* for the definition of the marginal probabilities  $P_{\text{indep}}$ . This factors the  $N$ -particle joint probability into  $N$  independent paths:

$$P_{\text{indep}}(X_t | \hat{X}_0) = \prod_{i=1}^N P_{\text{1D-CTMC}}(x_t^{(i)} | \hat{x}_0^{(i)})$$

The single-particle probability mass function for a symmetric integer random walk is highly tractable, typically governed by modified Bessel functions of the first kind  $I_v(z)$  (Skellam probabilities).

By substituting  $P_{\text{indep}}$  into the Anderson equation, we can derive the precise pseudo-rates used to tau-leap the generation backwards.

### 3.1 Reverse Mutation Rate

Particle  $i$  moves backwards from coordinate  $x$  to  $x + dx$ . The forward rate of arriving from  $x + dx$  is  $\gamma_{\text{mut}}\lambda(t)$ . The reverse rate evaluates to:

$$R_{\text{mut}}(i) = \gamma_{\text{mut}}\lambda(t) \frac{P_{\text{indep}}(x + dx | \hat{x}_0^{(i)})}{P_{\text{indep}}(x | \hat{x}_0^{(i)})}$$

Because the probability factorizes, all other  $N - 1$  particles completely mathematically cancel out in the fraction. Particle  $i$  executes an independent gradient ascent towards its neural target.

### 3.2 Reverse Resampling Rate

This is the crux of the model. In reverse generation, a population clumped natively into a cell type cluster by the Dirichlet Process prior must dynamically separate into distinct cells to resolve the Two Moons distribution.

Suppose particles  $i$  and  $j$  are currently co-located as clones:  $x^{(i)} = x^{(j)}$ . The forward rate of particle  $i$  arriving at this cloned state from some arbitrary, distant coordinate  $z$  was  $\frac{\kappa}{N}K(z, x^{(j)})\lambda(t)$ .

The reverse rate at which particle  $i$  breaks off from the cluster and leaps directly to  $z$  is:

$$R_{\text{resample}}(i \rightarrow z) = \frac{\kappa}{N}K(z, x^{(j)})\lambda(t) \times \frac{P_{\text{indep}}(z | \hat{x}_0^{(i)})}{P_{\text{indep}}(x^{(i)} | \hat{x}_0^{(i)})}$$

This beautifully demonstrates that reverse interactions are **not neutered**. The particle splits from the clump and jumps to  $z$  perfectly conditionally weighting:

1. The raw mathematical necessity of  $z$  being within spatial bounds of the cluster (enforced by the spatial kernel  $K$  belonging exclusively to the  $Q_{Y \rightarrow X}$  forward rate).

2. The directive mapping pulling the specific cell  $i$  towards its targeted  $\hat{x}_0^{(i)}$  feature space (enforced precisely by the independent ratio conditional).

Thus, by approximating the conditional probabilities independently, we naturally harness localized spatial correlation whilst enabling tractable reverse-time generation.

## 4 Justification: Analogy to Nuclear Adjoint Monte Carlo

The independent pseudo-marginal approximation developed above is not merely a mathematical convenience; it natively parallels the rigorous **Adjoint Monte Carlo** methods utilized via the Feynman-Kac formalism to solve the backward Boltzmann neutron transport equation in nuclear engineering.

In nuclear shielding frameworks, simulating the exact backward flow from a sparse detector to a distributed reactor source requires analytical access to the global interacting forward neutron flux  $p_t(X)$ . Because neutrons undergo constant streaming (analogous to Moran mutation) and fission/absorption branching (analogous to Moran resampling), this forward interacting joint distribution is famously mathematically intractable.

To bypass this intractability to solve the backward equations, theoretical nuclear models construct a non-branching, coarse deterministic simplified solver (such as independent discrete-ordinates) to extract a baseline distribution track. This analytical baseline serves as an **Importance Function**. During the backward stochastic Monte Carlo simulation, the reverse-fission and reverse-scattering jump rates are strategically multiplied by the ratio of this Importance Function between the candidate and current states.

Our Option A architecture explicitly executes this exact physical theorem:

1. We replace the intractable interacting particle forward probability with the analytically exact  $\kappa = 0$  independent state equations (Skellam/Bessel functions).
2. We utilize the ratio of these *Independent Pseudo-Marginals*—firmly anchored by the neural network’s detector target prediction  $\hat{X}_0$ —to act identically as the Importance Function gradient.

By injecting this ratio directly into the exact  $Q_{Y \rightarrow X}(t)$  branching mechanisms, we dynamically bias the backward tau-leaping generation without destroying the native non-linear particle coupling. If strict simulation parameters validate this foundational theorem in subatomic physics, it theoretically justifies our exact mathematical construction for generative population mapping.

## 5 Algorithmic Enhancements for Generative Stability

To enforce structural rigidity and alleviate pathological boundary accumulation during generative inference, three critical mathematical extensions have been structurally bound to the reverse tau-leaping simulator.

### 5.1 Toroidal Boundary Topologies

Because standard random walk clipping bounds artificially lock particle states at grid edges (forming high-density sticky boundaries asymptotically as  $t \rightarrow 1$ ), the DP prior noise structure breaks. To rectify this, the data coordinates are explicitly mapped to a modulo Torus:

$$X_t = (X_{t-\Delta t} + dX) \pmod{g_{max} + 1} \quad (1)$$

Respectively, spatial distance metrics governing pairwise resampling interactions and Skellam probabilities map directly to the absolute shortest cyclical gap bridging the Torus.

## 5.2 Pseudo Classifier-Free Guidance (Score Extrapolation)

To mitigate early-generation regression blurring where the un-guided neural network predicts fuzzy spatial interpolation natively minimizing MSE loss, an explicit temperature coefficient exponentially pushes jump rates into high-density spikes cleanly mapping discrete continuous-time Classifier-Free Extrapolations:

$$R_{mut} \propto \left( \frac{P_{\text{indep}}(X_{cand} | \hat{X}_0)}{P_{\text{indep}}(X_{current} | \hat{X}_0)} \right)^w \quad (2)$$

Using an extrapolated weight  $w > 1$  strictly isolates prediction structures precisely mirroring the exact temperature scaling methodology found natively in unbounded generalized diffusion mechanics.

## 5.3 Interaction Decay (Moran Conditional Bridging)

A strictly localized spatial resampling kernel mathematically dictates particles recursively anchor towards clones. When structural predictions carry any latent regression noise nearby paths deviate wildly. To precisely clamp inference onto target analytical boundaries without destroying the entire interactive tree structure, the scaling parameter  $\kappa$  linearly decays to absolute zero bounds at generation endpoints:

$$\kappa(t) = \kappa_{max} \cdot t \quad (3)$$

At  $t \rightarrow 0$ ,  $R_{jump} \rightarrow 0$  locking trajectory spans explicitly strictly mirroring the mathematically exact deterministic conditional Skellam bridge configurations organically!